

Week 1: From Process Prediction to Decisions

Define a process decision problem, build a reproducible baseline, and check the selected operating point with a reference CSTR model.

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Learning objectives

After this lesson, you should be able to write the inputs, outputs, fixed context, units, and operating domain of a surrogate; distinguish training from operational optimization; construct a steady-state and a dynamic CSTR example; and check whether a surrogate-selected operating point satisfies the reference model's constraints.

Suggested study time: 90 minutes for the notes and derivations, 60–90 minutes for the lab, and 2 hours for the exercises. The **syllabus** explains how this lesson connects to the rest of the course.

1. Begin with the decision

Suppose a reactor operator wants high conversion at a low operating cost. Predicting conversion at a proposed temperature is one task. Choosing temperature and residence time while meeting a conversion requirement is another task. The second task repeatedly queries the model in regions favored by the optimizer, which may differ from a typical held-out test sample.

Write four items before selecting a network: what can be changed, what is fixed for this decision, what must be predicted, and what counts as a satisfactory operating point. A small prediction error averaged across the domain does not tell us whether a constraint is satisfied at the selected point.

Role	Meaning	Example in this lesson
Decision variables	Quantities selected by the optimization	Reactor temperature T and residence time τ
Fixed context	Known quantities held fixed for one solve	Feed concentration set to 1.2 mol/L for the decision comparison
Model outputs	Quantities predicted for a proposed input	Outlet concentration and conversion
State	A quantity that evolves in a dynamic model	Reactor concentration
Domain	Conditions over which the model is evaluated	Stated bounds on temperature, residence time, and feed concentration

Training adjusts model parameters using data. Operational optimization holds the trained parameters fixed and changes the decision variables. These are different optimization problems.

2. A reference process we can check

Use an ideal, perfectly mixed, constant-volume, isothermal CSTR with a first-order reaction $A \rightarrow B$, constant density, and no B in the feed. For each operating condition, reactor temperature is externally maintained. We model the concentration dynamics and omit an energy balance. Temperature changes the reaction rate, but the heat-duty requirements are not calculated.

$$\frac{dC_A}{dt} = \frac{C_{Af} - C_A}{\tau} - k(T)C_A$$

Here concentrations are in mol/L, time and residence time in minutes, T in kelvin, and k in inverse minutes. The flow-to-volume ratio is the reciprocal of residence time. The inlet–outlet term and the reaction term therefore both have units of mol/(L·min).

The educational rate law is:

$$k(T) = k_{\text{ref}} \exp \left[\beta \left(\frac{1}{T_{\text{ref}}} - \frac{1}{T} \right) \right]$$

Parameter or input	Value or range	Interpretation
Reference rate constant	0.2 per minute	Rate at the reference temperature
Reference temperature	330 K	Rate-law reference
$\beta = E/R$	6000 K	Synthetic temperature-sensitivity parameter
Temperature	300–360 K	Sampling and decision domain
Residence time	1–10 min	Sampling and decision domain
Feed A concentration	0.8–1.5 mol/L	Sampling domain; fixed to 1.2 for optimization

These are synthetic teaching parameters, not fitted data from a real reactor or a published case study. Our simulator is the reference model for this exercise, not a claim of plant fidelity.

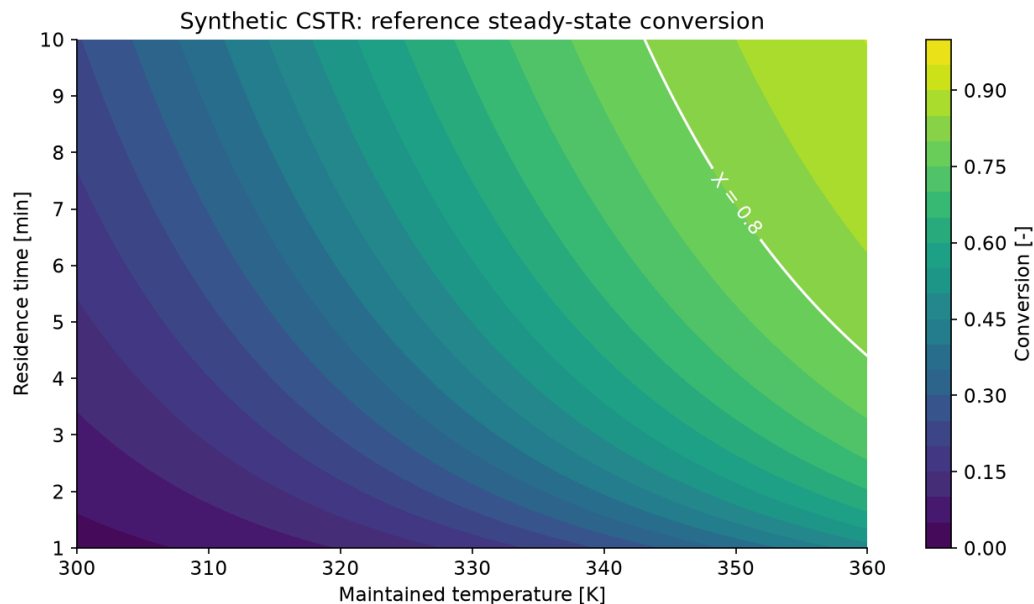
3. Derive the steady-state map

Set the time derivative to zero, multiply by residence time, and collect the terms containing outlet concentration. The resulting mapping and conversion are:

$$C_A = \frac{C_{Af}}{1 + k(T)\tau}, \quad X = 1 - \frac{C_A}{C_{Af}} = \frac{k(T)\tau}{1 + k(T)\tau}$$

This gives a three-input, two-output map: temperature, residence time, and feed concentration map to outlet concentration and conversion. Conversion is independent of feed concentration only under this particular first-order, isothermal model. Do not carry that conclusion to arbitrary kinetics or a nonisothermal reactor.

A worked check: at 330 K and 5 min, the product $k\tau$ is 1, so conversion is 0.5 and outlet A concentration is half the inlet concentration. For the 1.2 mol/L feed, the outlet is 0.6 mol/L. Both a simulator and a surrogate implementation should pass this basic reference check.



Reference-model conversion. The contour marks the conversion requirement used in the decision exercise.

4. State the optimization problem

Use an illustrative normalized cost index rather than an industrial economic model:

$$J(T, \tau) = \frac{T - 300\text{K}}{20\text{K}} + \frac{\tau}{10\text{min}}$$

The index penalizes higher temperature and longer residence time. It is dimensionless and is not a heat-duty, profit, or currency calculation. At a fixed feed concentration of 1.2 mol/L, minimize this index over the stated bounds subject to predicted conversion of at least 0.8. Also require predicted concentration to lie between zero and the feed concentration, and predicted conversion not to exceed one.

For the reference map, conversion of at least 0.8 is equivalent to $k\tau$ of at least 4. This follows by multiplying the conversion inequality by the positive denominator. It gives a useful independent check on the feasible region.

Replacing the reference map with a surrogate changes the feasible set. Solving that surrogate problem accurately cannot establish feasibility for the reference process. Run the selected temperature and residence time through the reference model and report the conversion shortfall, if any.

5. Design data for the intended use

The lab generates 400 training, 120 validation, and 160 test points independently inside the stated box, using a fixed random seed. Fit input scaling using the training definition, fit model coefficients using training data only, use validation for model choices, and reserve the test set for reporting. The supplied baseline uses fixed domain scaling and a preselected quadratic feature set; it does not tune on test results.

The two-output baseline is ordinary least-squares regression on ten features: a constant, three scaled inputs, their squares, and their pairwise products. This keeps the first lesson focused on the decision workflow. It is not a ReLU model or a MILP embedding; later lessons replace it with neural models and derive their formulations.

For measured data, split by batch, experiment, campaign, or time when those groups share information. For dynamic data, overlapping windows from the same trajectory can leak information across a random row split. Entire trajectories or independent operating campaigns should be held out when evaluating generalization to new trajectories.

Being inside an input box is not sufficient evidence of reliable interpolation. Inspect sampling density, operating regimes, and the error near the constraint boundary. The lab's independent box samples are a controlled teaching design, not a substitute for this analysis on real data.

6. Prediction error and decision error

Report concentration RMSE in mol/L and conversion RMSE as a dimensionless fraction. Also report the consistency residual relating the two outputs: predicted outlet concentration plus feed concentration times predicted conversion minus feed concentration. A multi-output fit can disagree with this identity.

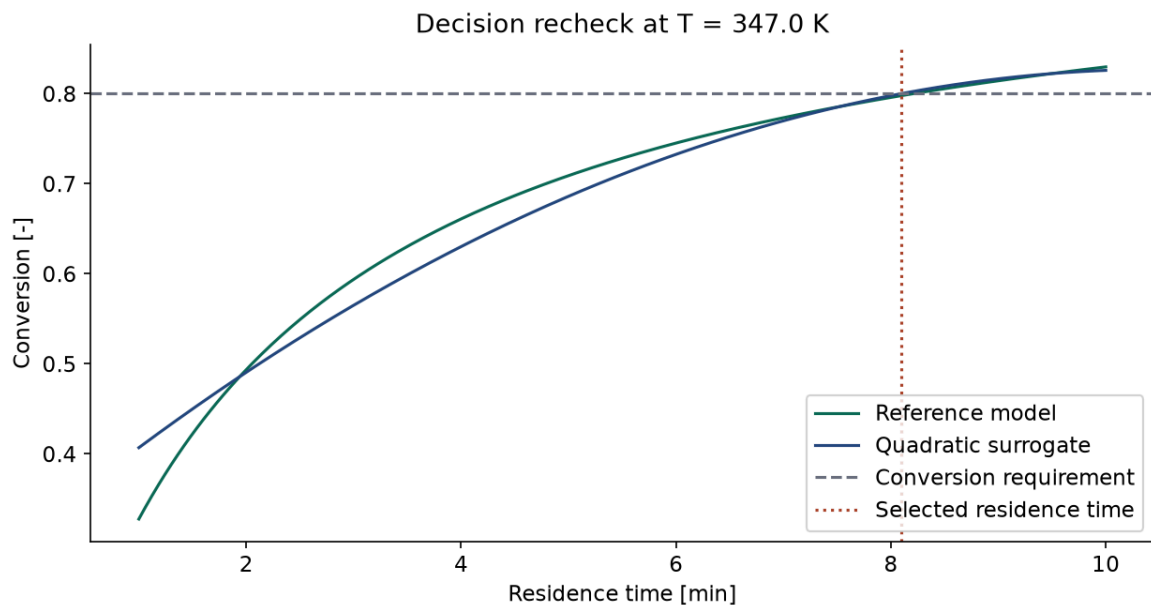
The decision comparison enumerates the same grid for the surrogate and reference maps: temperature steps of 0.5 K and residence-time steps of 0.1 min. It finds the lowest-cost feasible candidate on that finite grid. It does not certify a continuous global optimum.

The values below are generated by the downloadable lab with seed 42. They describe this synthetic example, not paper results.

Check	Result
Held-out concentration RMSE	0.032623 mol/L
Held-out conversion RMSE	0.025608
Selected surrogate decision	347.000000 K; 8.100000 min
Predicted conversion at that decision	0.800144
Reference conversion at that decision	0.797893
Reference conversion shortfall below 0.8	0.002107
Reference cost of the surrogate-selected decision	3.160000

Check	Result
Reference-model feasible grid benchmark cost	3.155000

If the selected decision is infeasible under the reference model, a lower cost is not an economic improvement. A meaningful cost comparison requires reference feasibility. The lab reports feasibility first and only reports a feasible decision cost gap when the selected decision passes the reference check.



A local check near the selected decision. Error close to the constraint boundary can change the feasible operating choices.

7. Connect vectors to time series

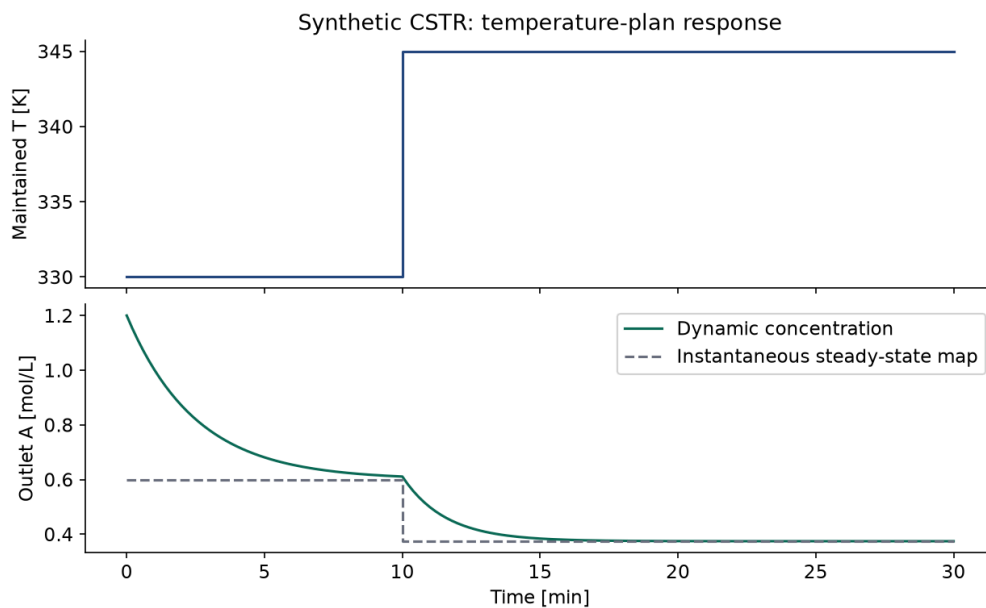
A steady-state output has no time index. A dynamic output depends on the initial state and the history of control inputs and disturbances. In a planning problem, the model must also receive the future control sequence being considered; predicting from historical measurements alone is insufficient to compare arbitrary future control plans.

The dynamic lab holds the feed concentration and residence time fixed and changes the maintained temperature from 330 to 345 K at 10 min. Within each constant-input interval, the concentration balance has an analytic update:

$$C_A(t + \Delta t) = C_{Ass} + \left[C_A(t) - C_{Ass} \right] \exp \left[- \left(\frac{1}{\tau} + k(T) \right) \Delta t \right]$$

The steady concentration in this expression is recomputed for the input of each interval. The update is exact for the teaching ODE with piecewise-constant inputs, subject to floating-point arithmetic. It is not a time discretization of an arbitrary nonlinear process.

The code records input values on intervals and concentration on interval boundaries. At the step time, the concentration is continuous and then approaches the new steady value. This indexing convention will matter when we train sequence models in Week 3.



Dynamic reference response. An instantaneous steady-state mapping would miss the transient concentration trajectory.

8. Run the lab

Download the **notebook** or the **standalone script**, plus **requirements.txt**. The notebook includes English and Korean instructions and all required model code. It does not need an external dataset or an API key.

```
python -m pip install -r requirements.txt
python week01_lab.py --output-dir week01-results
```

The script creates the synthetic dataset, metrics JSON, dynamic trajectory CSV, and three figures inside the chosen output directory. The **published dataset** and **recorded results** are also available for comparison. Small numerical differences can occur across library versions; the operating grid and random seed are fixed.

9. Exercises and submission

1. Derive the steady-state concentration and check its units. Explain why increasing either temperature or residence time increases conversion under this model.
2. Derive the condition $\kappa\tau \geq 4$ for the conversion target. Check the reference feasible region without fitting a model.
3. Re-run the lab with a different training seed. Compare test RMSE, the selected operating point, and reference feasibility. Does better RMSE always select a better decision?

4. Tighten the temperature domain to 330–350 K. Explain how the feasible set and the sampled training domain should change together.
5. Change the temperature step time. Describe which input intervals and output boundary times change. Explain why a steady-state predictor cannot represent the transient.
6. Write a one-page problem specification stating the decision variables, fixed context, state, outputs, units, domain, objective, constraints, data split, and revalidation rule.

Reference checks: at 330 K and 5 min, conversion is 0.5; the conversion target requires $k\tau \geq 4$; increasing T increases k for positive β . The other exercises require your computed results and interpretation, including failed constraints when they occur.

Reading and next lesson

Read selected parts of Boyd and Vandenberghe's **Chapters 2–4** on domains, constraints, convex functions, and optimization problems. Read the **OMLT introduction** for the role of trained models in larger optimization problems. The CSTR equations and synthetic parameters above are original teaching material derived from the stated balance assumptions.

Week 2 replaces this simple baseline with a multi-output MLP. Keep this lesson's data contract and reference checks when changing the model.